

Robustness, Cascades and Bus Recovery in the Transperth Network

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Abstract

We model robustness, overload cascades and bus recovery on an audited 86-station Transperth railway. Targeted removal fragments the network sooner than random station loss. Local load redistribution produces abrupt secondary failure, whereas shortest-path recomputation on the rail tree contains overload by dropping unreachable journeys. Bus links can restore service while increasing rail failures; with a two-link policy after an Airport Line closure, served terminal pairs fall from 93.11% to 66.68%. Sampling and grid checks qualify the tolerance thresholds, and the finite avalanche samples do not support a power-law claim. These findings concern specified routing and capacity proxies, rather than measured passenger behaviour or operational deployment advice.

1 Introduction

1.1 Background and research aims

A station closure can sever a railway corridor before any further facility fails. Substitute buses may reconnect journeys, but the paths they restore can also concentrate traffic at surviving rail facilities. We investigate these competing effects in the Transperth network, distinguishing infrastructure connectivity, secondary overload failures and the fraction of terminal pairs that remain served. Fewer open rail facilities need not imply less service when another transport layer is available.

Albert et al. (2000) established the distinction between random errors and targeted removal in heterogeneous networks. Their scale-free-network results motivate a comparison of failure strategies here; they do not establish that Transperth has the same degree distribution or resilience. Our audited railway has 86 mapped stations and 85 adjacent segments and is a tree. It therefore lacks alternative rail paths around a removed station. Its inputs represent Monday 5 October 2026, 07:00–09:00 in Australia/Perth, with 271 selected rail trips and 4,324 selected bus trips. These are scheduled services, not measured passenger journeys. Showgrounds remains a mapped station although it has no weekday stop event in this window.

Motter and Lai (2002) showed how a local attack can cause further overload failures when shortest-path loads are recomputed against capacities fixed from the intact network. We use that capacity construction and compare it with a local redistribution scenario in which a failed facility forwards its carried load to surviving neighbours. These rules represent different assumptions about disrupted travel. Their comparison tests whether an apparent tolerance transition depends on the update mechanism. In the bus layer, restored paths can increase surviving rail loads even while improving terminal reachability. Apparent avalanche scaling requires a separate statistical test: following Clauset et al. (2009), a visual slope alone is not evidence for a power law.

Our research questions are: **RQ1**, how do random and targeted removal fragment the railway, and which station characteristics identify bottlenecks? **RQ2**, how do capacity tolerance and load-update rules affect the size and duration of secondary failures? **RQ3**, when do bus backup links improve served travel, and can their rerouting aggravate a rail cascade? **RQ4**, how sensitive are these conclusions to sampling, tolerance-grid resolution and the assumed load/capacity model, and is avalanche scaling statistically supported?

1.2 Contribution

Our contribution is a reproducible transport case study that joins an audited GTFS-derived topology to removal, overload and recovery experiments. We expand express-service traversals over mapped adjacent segments rather than treating skipped stops as shortcuts. We separate persistent passenger terminals from closable rail facilities, retain evidence for substitute links, and compare infrastructure damage with service outcomes. Paired scenarios and bootstrap intervals expose uncertainty without treating scheduled departures as observed demand. The capacity law and robustness questions come from existing work; the audited Transperth construction, explicit comparison of load-update assumptions, and evidence-linked recovery analysis are the team’s extension. The implementation is network simulation; the unit’s taught agent-based models are not reused.

2 Model specification

The model represents the transport network as a graph of entities that carry a load and fail when the load exceeds a fixed capacity. Entity i has an open or closed status $s_i(t)$ and a carried load $L_i(t)$, and it interacts with its neighbours $\mathcal{N}_i$. With the state $x_i(t) = (s_i(t), L_i(t))$, the generic update rule is

$$x_i(t+1) = f(x_i(t), \{x_j(t)\}_{j \in \mathcal{N}_i}; \theta), \quad (1)$$

where θ collects the capacity, tolerance, redistribution and rerouting parameters. In the transport case f closes every open entity above capacity, removes the round’s overloads together, and forwards their load to surviving neighbours, following the synchronous rule and capacity construction of Motter and Lai (2002) with the local-redistribution variant below. A round is an algorithmic update and carries no wall-clock meaning.

2.1 Components, loads and failure

The rail layer is an undirected simple tree of 86 mapped stations and 85 verified adjacent segments, frozen for 5 October 2026, 07:00 to 09:00 Australia/Perth. The multilayer experiments split each station into a persistent terminal $T:i$ and a closable rail facility $R:i$ with a one-minute access edge. Rail edges take $t_e = \max(0.1, d_e/(1000v) \cdot 60)$ minutes at an assumed $v = 40$ km/h; backup edges join terminals with effective times including walking and waiting, with the four manual pairs in `backup_edges.csv` as ground truth and GTFS candidates extending the pool; a five-minute transfer joins the Perth and Perth Underground terminals. Standby buses activate immediately after the initial closure with unlimited capacity in the baseline scenario. Multilayer loads are terminal-subset betweenness with terminals as sources and targets, and served travel keeps the intact terminal-pair denominator.

The initial load of station i is its unnormalised betweenness on the intact graph, $L0_i = \sum \sigma_{st}(i)/\sigma_{st}$, summed over station pairs with $s \neq i \neq t$, where the ratio is the share of shortest paths from s to t through i . The values stay unnormalised because the graph shrinks during a cascade: normalised loads would rise with the shrinking node set and rounds would not be comparable. The load modes are `betweenness` as above; `betweenness_freq` with edge length $1/\text{trips}$; `betweenness_plus_trips` with 0.1 trips served _{i} added to give leaf stations a positive

baseline; and **demand**, the AM-peak scheduled departure count **am_peak_stops**, a proxy for boardings rather than observed passengers.

Capacities follow

$$C_i = (1 + \alpha)K_i, \quad \alpha \geq 0, \quad (2)$$

with default reference capacity $K_i = L0_i$, fixed from the intact graph. A station with $L0_i = 0$ gets $C_i = 0$ and can still fail under positive redistributed load in the static mode. The demand experiment uses $K_i = cf_i$, where $f_i = \text{trips_served}_i$ and $c = \max_i L0_i/f_i$ is the smallest uniform scaling that keeps $K_i \geq L0_i$ on the frozen graph; the scale c is a scenario assumption recorded with the demand results.

At each round every surviving station with $L_i > C_i + \delta$ fails, where the numerical tolerance δ defaults to 10^{-12} ; all overloads of a round are collected before any station is removed, and the cascade stops at the first clear round. The first failure is an explicit **target** when one is named, otherwise the **load**, **degree** or **random** trigger takes the largest initial load, the highest degree with station-ID ties, or a uniform seeded draw. A line closure replaces the trigger with one initial batch, removed before the first check and excluded from the secondary avalanche size. In the static mode each closed station forwards its load equally over surviving neighbours or in proportion to their capacities, falling back to equal shares when all have zero capacity; a station with no survivor removes its load, and otherwise load is conserved. The dynamic mode recomputes loads on the surviving graph each round and ignores redistribution; on a tree this cannot raise a survivor’s betweenness, so containment reflects dropped journeys rather than headroom. The containment tolerance α^* is the smallest sampled α at which the maximum-load trigger causes no secondary failure.

2.2 Assumptions and initialisation

All facilities start open; loads and capacities come from the intact graph, and the trigger or initial batch is applied before the first overload check. The model assumes undirected adjacent links, scheduled services as load proxies, hop distances for load concentration, minute travel times only in the multilayer layer, symmetric effective bus times, immediate unlimited standby, and a single service date and window. An empty initial batch closes nothing; an α of zero adds no headroom and can fail a station immediately; duplicate backup endpoints keep the fastest connection; zero capacity does not exempt a station from overload checks.

The defaults in `src/transperth/config.py` are $\alpha = 0.2$, master seed 0, tolerance 10^{-12} , and the 81-point grid $\alpha = 0, 0.025, \dots, 2$. The tolerance only guards floating-point comparisons. The grid starts at zero and reaches twice the $L0$ reference; it brackets the sampled containment tolerances near 0.5 and 0.75, and the 0.025 step resolves the abrupt static transition, as the uncertainty experiment confirms.

3 Experimental design

Five runner families produce the evidence, each from one command in the repository root. They load the frozen graph, apply the model of Section 2, and write their tables and figures under **results/** and **figures/**. Table 1 lists each family’s scenarios, runner defaults and outcome measures.

All runs use master seed 0 and derived child seeds, so paired conditions share random draws while repeated draws remain separate observations. Each runner writes one row per replicate through `experiments.save_table`, with a `.meta.json` sidecar holding the seed, parameters, package versions and SHA-256 hashes of inputs and code, plus a family manifest that hashes the artifacts; `-quick` variants are smoke tests in separate directories.

The pipeline is runner, stored rows, statistics, report. The runners call the packaged engines and store raw per-seed rows under **results/**<family>; the P1.2 helpers summarise them

Table 1: Scenarios, runner defaults and outcome measures.

Command		Scenarios and runner defaults	Outcome measures
RQ1: percolation	make	Random and targeted node and edge removal ranked by degree, betweenness and demand flow; fractions 0.00 to 0.90 in 0.01 steps plus 1.00, 100 seeds each; every station once as trigger over 81 tolerances ($\alpha = 0$ to 2, step 0.025), static capacity rule	Seed-mean GCC fraction and isolates; susceptibility peak; GCC = 0.5 crossing; containment tolerance; predictor strength and rank stability
RQ2: cascade	make	Static sweep $\alpha = 0$ to 2 in 0.025 steps with equal and capacity rules; maximum-load, maximum-degree and 300 seeded random triggers. A reduced 10-point grid adds the frequency and demand load modes and dynamic rerouting. 2,000 random triggers at $\alpha = 0.1, 0.15, 0.2, 0.3$; line closures as one batch	Failed fraction; avalanche size, duration and span; containment α^* ; paired rule, load-mode and rerouting differences
RQ3: recovery	make	21 closures: 8 exclusive-station lines, 8 interchanges, 5 seeded random sets of 5. Five strategies with a 2-link budget over 4 manual and 266 GTFS-derived candidate pairs; layered cascade $\alpha = 0$ to 2 in 0.05 steps; focus checks at 0 and 0.2	Served OD fraction and endpoint-sum demand over 3,655 terminal pairs; secondary rail closures and recovery rounds; travel-time penalty conditional on reachable pairs
RQ4: demand	make	Topology (betweenness loads, $K = L0$) against demand (am_peak_stops loads, frequency-scaled K); 81 tolerances from 0 to 2 in 0.025 steps; maximum-load trigger; capacity rule	Load and rank changes; Spearman correlation; top-5, top-10 and top-20 overlap; containment-tolerance shift
RQ4: uncertainty	make	1,000 random draws per removal fraction (0 to 1, step 0.05); 95% percentile bootstrap with 2,000 resamples; nested seed prefixes 25 to 2,000; alpha grids from 0.5 to 0.025; power-law test with 500 refitted simulations at $\alpha = 0.1, 0.15, 0.2, 0.3$	Pointwise interval half widths; seed convergence; grid-induced containment shift; tail eligibility and rejection probability

as means and percentile intervals; **make figures** checks the tables against their sidecars and manifests, then writes the report figures and the macros in **sections/00-results-values.tex**; and Section 4 cites those frozen files. Notebooks 01 to 04 (issues P3.1 to P3.4) re-execute the same calls interactively. **make check** runs the test suite and **make reproduce** chains **data**, **check**, **figures** and **notebooks**; each family regenerates with its command in Table 1. **run_uncertainty.py** verifies the committed P2.2 manifest before running, so P2.2 is regenerated first if its inputs change.

4 Results

4.1 RQ1: targeted removal exposes corridor bottlenecks

Figure 1a shows earlier fragmentation under targeted station removal. The interpolated removal fraction at which the mean largest component reaches half of the intact 86 stations is 6.32% for random removal, 1.68% for a fixed degree ranking and 1.24% for a fixed betweenness ranking. These crossing estimates are distinct from the corresponding second-largest-component susceptibility peaks at 6%, 2% and 1%. They also depend on integer station-count rounding and grid resolution. On this small tree, neither diagnostic establishes a thermodynamic phase transition. Betweenness identifies corridor-separating positions that degree alone cannot distinguish: many internal stations have degree two but sit between different amounts of the network.

4.2 RQ2: tolerance depends on the load-update assumption

For the maximum-load trigger, Oats Street, static capacity-weighted redistribution at $\alpha = 0.2$ closes all 86 stations, including the trigger, in 31 secondary rounds. On the 0.025 grid, the first sampled tolerance containing the closure is $\alpha = 0.525$; equal redistribution requires $\alpha = 0.625$ (Figure 1b). These are sampled thresholds for the specified trigger and rule, not universal station-capacity requirements. Dynamic unweighted betweenness produces no secondary failures on the frozen tree: deletion cannot create a new rail route, so surviving path loads cannot

increase. The resulting disconnected graph can still lose service. The 2,000 random-trigger observations include zero secondary failures, and rounds within one cascade are not separate trials. The observed size distribution is concentrated at a few finite outcomes (Figure 3b).

4.3 RQ3: service recovery and rail damage can move in opposite directions

After a Bayswater closure at $\alpha = 0.2$, the rail-only scenario serves 67.00% of the original 3,655 terminal pairs. Full candidate bus standby serves 100.00%, but closes two rail facilities instead of one (Figure 2a). At the first load check, Fremantle’s load-to-capacity ratio rises from 0.68 without buses to 1.31 with the full pool (Figure 2b). It exceeds the fixed rail-only capacity and fails next. Bus paths still keep its terminal and the other terminals connected. At $\alpha = 1.0$, the full-pool Bayswater case serves all pairs with only the trigger closed. Additional capacity can therefore contain this rerouting overload without being responsible for the restored paths themselves.

The complete candidate pool does not worsen served service in the stored scan; a budgeted strategy can. After the three exclusive Airport Line facilities close at $\alpha = 0.2$, the existing-bus policy selects two links and additionally closes West Leederville. Served terminal pairs fall from 93.11% with no backup to 66.68%. Endpoint-weighted served demand likewise falls from 94.50% to 69.65% (Table 2). These weights are sums of scheduled stop counts at pair endpoints, not observed OD demand. Travel-time penalties are conditional on reachable pairs; comparing their means alone could mistake loss of long journeys for faster service.

4.4 RQ4: numerical uncertainty and model sensitivity

The tested nested-prefix criteria recommend 1000 random trials: the worst normalised mean change is 0.0030 and the largest pointwise 95% interval half width is 0.0220, below the chosen 0.02 and 0.03 tolerances (Figure 3a). These criteria apply to the stored conditions, not every transport model. Coarsening the capacity-rule alpha grid from 0.025 to 0.1 shifts first sampled containment from 0.525 to 0.600. The demand/frequency-capacity scenario gives 0.750; it changes the load proxy, capacity reference and maximum-load trigger, so it is a model comparison rather than an isolated effect of passenger demand. Frequency weighting alone leaves unique tree paths unchanged.

The bounded decreasing power-law test follows maximum-likelihood fitting, cutoff selection and refitted simulation, rather than log-log regression (Clauset et al., 2009). At static alpha 0.1–0.2 there are only two distinct positive secondary sizes, making the tail ineligible under the three-size requirement. At 0.3 the fitted hypothesis is rejected ($p = 0.002$, 500 refitted simulations), with its exponent at the lower search boundary. Dynamic samples contain only zeros. The experiments therefore do not support a power-law avalanche claim.

5 Discussion and conclusions

The tree topology explains RQ1’s vulnerability: removing a station can cut the only rail path between corridors. Extra capacity cannot restore a deleted edge. RQ2 instead concerns whether surviving facilities acquire further overloads. Static redistribution keeps forwarding a proxy quantity locally; dynamic betweenness removes contributions from unreachable pairs. Interpreting the dynamic result as uninterrupted operation would confuse disappearing load with preserved service. Likewise, the static all-station cascade is conditional on the forwarding rule, not evidence that a single real closure would destroy Perth’s railway.

RQ3 shows why recovery needs both service and facility metrics. Backup paths can reconnect terminals while increasing load at particular rail facilities; the Fremantle case supplies a traced overload mechanism. Enough substitute connectivity can maintain travel despite that extra closure, whereas a limited selection can leave the network worse after secondary failures. Policies

should therefore be compared after the cascade stabilises, with a common intact OD denominator and the reachable population reported beside travel times. Our two-link budget counts selected connections, not physical buses or fleet cost.

Uncertainty remains broader than the Monte Carlo intervals in RQ4. We study one service date and window, mapped undirected links, approximate rail speeds, proxy capacities, symmetric effective bus times and immediate unlimited bus standby. The model omits measured boarding/alighting, vehicle occupancy, directional timetables, congestion, dispatch delay and finite substitute capacity. Terminal-pair reachability is not proof of a feasible scheduled journey. The finite 86-station support also limits avalanche-tail inference; the rejected bounded hypothesis cannot establish the absence of scaling in a larger or different transport system.

The most useful next tests would introduce observed OD demand and vehicle capacity, directional time-dependent routing, delayed finite bus deployment, and several service-day snapshots. Enumerating all 86 triggers would remove trigger-sampling error, and intervals across dates would cover a different uncertainty. The conclusions are narrower: targeted corridor removal is structurally damaging; overload containment depends on the update rule and tolerance grid; and substitute connectivity can improve service while aggravating rail overload. These are reproducible mechanisms to test against operational evidence, rather than a recommendation to deploy a particular bus policy.

Selected figures

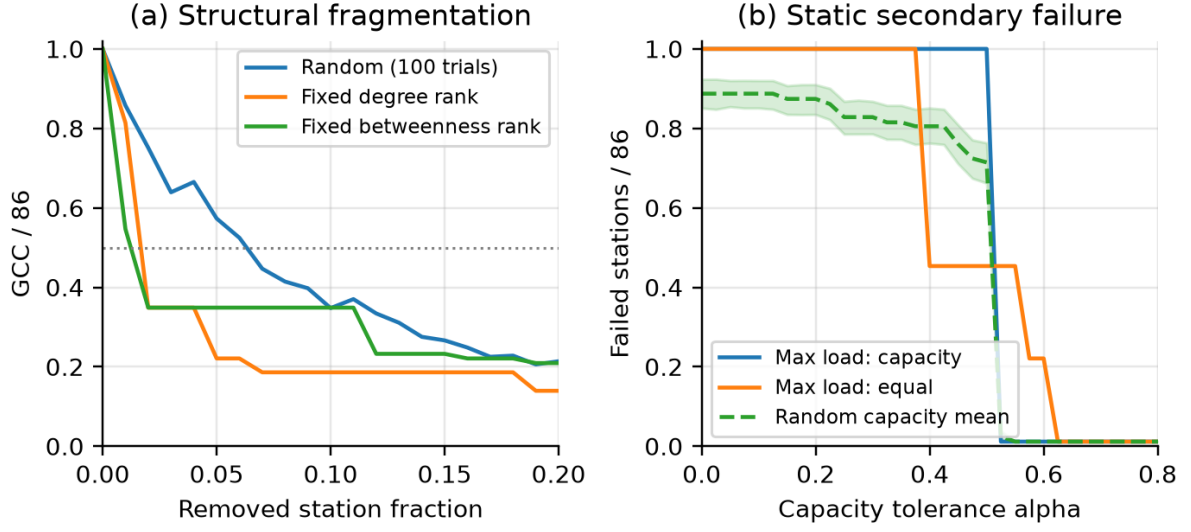


Figure 1: Frozen 86-station railway. (a) Station-removal curves, with the horizontal $\text{GCC}/86=0.5$ diagnostic. The random mean uses 100 trials per fraction; target orders are fixed from the intact graph. (b) Static unweighted-betweenness cascades. Targeted curves use the maximum-load station; the random capacity-rule mean uses 300 whole trigger trials per alpha and pointwise 95% bootstrap intervals with 2,000 resamples. Failed fractions include the trigger. Sources: P2.1 percolation tables, P2.2 runs and P2.5 cascade intervals; selected by `report/generate_figures.py`.

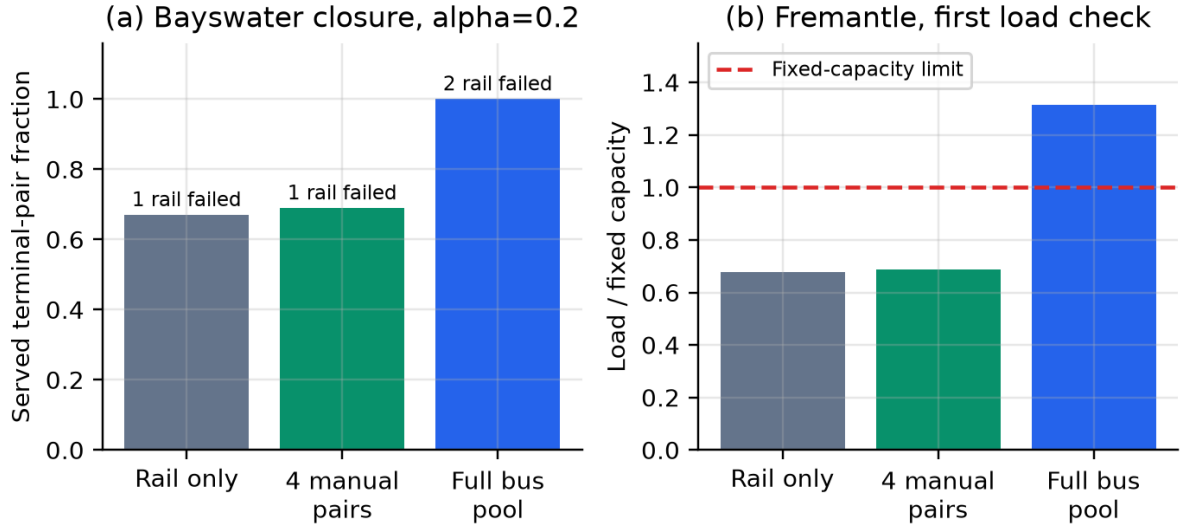


Figure 2: Bayswater closure at $\alpha = 0.2$, layered dynamic routing with capacities fixed from the intact rail-only baseline. (a) Served terminal pairs over the original 3,655 pairs; annotations count failed rail facilities, including the trigger. (b) Fremantle's load/capacity ratio immediately after the trigger, before secondary closures. The full bus pool restores all pairs while pushing Fremantle above one. This is immediate unlimited standby, not a two-link budget. Sources: P2.3 bus-cascade scan and loads tables.

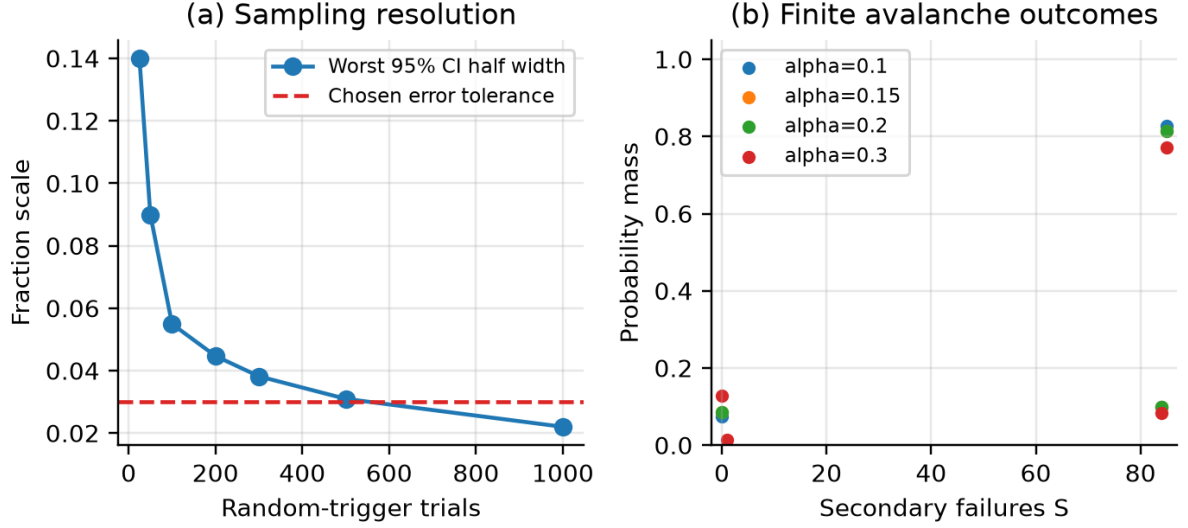


Figure 3: Uncertainty is conditional on the frozen graph and load model. (a) Worst pointwise 95% interval half width over the tested conditions with a common available prefix size. The 0.03 line is a chosen numerical tolerance; the accompanying mean-change criterion is checked separately. (b) Static capacity-rule probability masses from 2,000 random triggers at each alpha; zero secondary failures remain in the denominator and alphas are not pooled. Most mass lies at a few finite sizes, unsuitable evidence for broad scaling. Sources: P2.5 recommendations and P2.2 avalanche samples.

References

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- Aaron Clauset, Cosma Rohilla Shalizi, and M. E. J. Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009. doi: 10.1137/070710111.
- Adilson E. Motter and Ying-Cheng Lai. Cascade-based attacks on complex networks. *Physical Review E*, 66:065102, 2002. doi: 10.1103/PhysRevE.66.065102.

A Reproducibility and supplementary evidence

The repository is <https://github.com/CITS4403-Project/Project>. The live demonstration uses `notebooks/01_network.ipynb` and `notebooks/03_cascades.ipynb`, alongside the P3.2/P3.4 notebooks. These run from committed inputs and do not require a live timetable download. Full model definitions and raw-feed reconstruction are in `docs/model.md` and `docs/data.md`.

The RQ1 crossing table is `percolation_critical_fractions.csv` in `results/percolation/`; susceptibility peaks and GCC crossings occupy separate columns. RQ2 uses `results/cascade/runs.csv` and `avalanche_samples.csv`. RQ3 uses `results/recovery/bus_cascade_scan.csv`, `bus_cascade_loads.csv` and `strategy_comparison.csv`. RQ4 uses `results/uncertainty/recommendations.json`, `power_law_fits.json` and `results/demand/demand_alpha_star.csv`. Every source has a metadata sidecar. The selected-figure manifest additionally records table, sidecar, generator and output hashes.

From the repository root, `make figures` regenerates the three selected images and the numeric macros used in the abstract and results. Then run `make` inside `report/`, or use the documented Tectonic command in `report/README.md`. The P2 runners reproduce full experiments; the figure generator only reads their frozen outputs. Bootstrap intervals quantify pointwise Monte Carlo error; they do not cover different dates, measured demand or model misspecification.

Table 2: Airport Line exclusive-station closure, $\alpha = 0.2$. The policy budget is two selected links. Fractions use common intact denominators.

Policy	Secondary closures	Served OD (%)	Served weight (%)
No backup	0	93.11	94.50
Existing bus	1	66.68	69.65